

Inequalities with the Distributive Property

MA.8.AR.2.2

I can solve an inequality that has a bracket in it, and know when the symbol moves.

Lesson 5's brackets, meeting Lesson 8's reversal.

A negative can do two different things. Only one of them flips the symbol.

Warm-Up

On your own. Four minutes.

Same and different

What is the same about these two, and what is different?

$$-6(8 - x)$$

$$-30 \geq -6x$$

Notes

I. The same two methods as Lesson 5

- A. **Method 1. Distribute, then solve as usual.**
- B. **Method 2. Divide both sides first, then solve as usual.**
- C. Both give the same solution set. Neither is required.

Notes

II. Distributing a negative does NOT move the symbol

- A. **Distributing rewrites ONE side. Nothing was done to both sides.**
- B. $-6(8 - x)$ becomes $-48 + 6x$. The symbol stays exactly where it was.
- C. **You have not multiplied the inequality by anything. You have tidied one side of it.**

Notes

III. Dividing by a negative DOES move it

- A. **Dividing by a NEGATIVE is an operation on both sides, so the symbol reverses.**
- B. $-30 \geq -6(8 - x)$ divided by -6 becomes $5 \leq 8 - x$.
- C. **Count the divisions by a negative. That is how many times the symbol moves.**

Notes

IV. Check a solution SET by testing two numbers

- A. **Pick one number inside your answer and one outside it.**
- B. The inside one must make the original true. The outside one must make it false.
- C. **SWAPPED — inside false, outside true — means the symbol is reversed.**

Example 1

Which method, and does anything flip?

$$3(\mathbf{x} - 6) \geq 9$$

Answer both questions out loud before anybody writes.

Example 1

Both methods, and neither flips.

Method 1: distribute	Method 2: divide by 3
$3x - 18 \geq 9$	$x - 6 \geq 3$
$3x \geq 27$	$x \geq 9$
$x \geq 9$	

Example 1

Check with two numbers.

x = 10, inside	x = 0, outside
$3(10 - 6) = 12, \quad 12 \geq 9 \quad \checkmark$	$3(0 - 6) = -18, \quad -18 \geq 9 \quad \times$

Example 2

Now the multiplier is negative.

$$-3(\mathbf{x} - 6) \geq 9$$

Which method flips, and when?

Example 2

Two roads, one flip each, different moments.

Method 1: distribute first	Method 2: divide by -3 first
$-3x + 18 \geq 9$ no flip yet	$x - 6 \leq -3$ FLIPPED here
$-3x \geq -9$	$x \leq 3$
$x \leq 3$ FLIPPED here	

Example 2

Check with two numbers.

x = 0, inside	x = 10, outside
$-3(0 - 6) = 18, \quad 18 \geq 9 \quad \checkmark$	$-3(10 - 6) = -12, \quad -12 \geq 9 \quad \times$

Example 3

And the two sides the other way round.

$$9 \geq 3(x - 6)$$

The symbol looks backwards. Does anything flip?

Example 3

Nothing flips.

$$9 \geq 3(x - 6)$$

$$3 \geq x - 6 \quad \text{divided by POSITIVE 3}$$

$$9 \geq x, \text{ so } x \leq 9$$

Example 3

Three problems, three answers.

$3(\mathbf{x} - 6) \geq 9$	$-3(\mathbf{x} - 6) \geq 9$	$9 \geq 3(\mathbf{x} - 6)$
$\mathbf{x} \geq 9$	$\mathbf{x} \leq 3$	$\mathbf{x} \leq 9$
no flip	one flip	no flip

Whiteboards • Question 1 of 9

Boards up on three.

$$3(x - 6) \geq 9$$

Solve it.

A solution set. Say how many flips, and where.

Whiteboards • Question 1 of 9

Answer.

$$3(x - 6) \geq 9$$

divide by POSITIVE 3: $x - 6 \geq 3$

$$\textbf{Answer: } x \geq 9$$

Whiteboards • Question 2 of 9

Boards up on three.

$$-3(x - 6) \geq 9$$

Solve it.

A solution set. Say how many flips, and where.

Whiteboards • Question 2 of 9

Answer.

$$-3(x - 6) \geq 9$$

divide by -3 : $x - 6 \leq -3$

Answer: $x \leq 3$

Whiteboards • Question 3 of 9

Boards up on three.

$$3(-x - 6) \geq 9$$

Solve it.

A solution set. Say how many flips, and where.

Whiteboards · Question 3 of 9

Answer.

$$3(-x - 6) \geq 9$$

$$-x - 6 \geq 3, \text{ then } -x \geq 9$$

$$\textbf{Answer: } x \leq -9$$

Whiteboards • Question 4 of 9

Boards up on three.

$$3(x - 6) \geq -9$$

Solve it.

A solution set. Say how many flips, and where.

Whiteboards · Question 4 of 9

Answer.

$$3(x - 6) \geq -9$$

divide by POSITIVE 3: $x - 6 \geq -3$

$$\textbf{Answer: } x \geq 3$$

Whiteboards • Question 5 of 9

Boards up on three.

$$-3(x - 6) \geq -9$$

Solve it.

A solution set. Say how many flips, and where.

Whiteboards • Question 5 of 9

Answer.

$$-3(x - 6) \geq -9$$

divide by -3 : $x - 6 \leq 3$

Answer: $x \leq 9$

Whiteboards • Question 6 of 9

Boards up on three.

$$9 \geq 3(x - 6)$$

Solve it.

A solution set. Say how many flips, and where.

Whiteboards · Question 6 of 9

Answer.

$$9 \geq 3(x - 6)$$

$$3 \geq x - 6, \text{ then } 9 \geq x$$

$$\textbf{Answer: } x \leq 9$$

Whiteboards • Question 7 of 9

Boards up on three.

$$3(7 - 2h) > 15$$

Solve it.

A solution set. Say how many flips, and where.

Whiteboards • Question 7 of 9

Answer.

$$3(7 - 2h) > 15$$

divide by 3: $7 - 2h > 5$, then $-2h > -2$

Answer: $h < 1$ one flip, at the end

Whiteboards • Question 8 of 9

Boards up on three.

$$21.6 \leq 7.2(12 + x)$$

Solve it.

A solution set. Say how many flips, and where.

Whiteboards · Question 8 of 9

Answer.

$$21.6 \leq 7.2(12 + x)$$

divide by 7.2: $3 \leq 12 + x$

Answer: $x \geq -9$ no flips

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

$$-(5/2)(\frac{3}{4}y + 6) < -22\frac{1}{2}$$

Write your answer in sentences.

This one is written work. Say how many flips, and where each one happened.

Whiteboards • Question 9 of 9

Answer.

$$-(5/2)(\frac{3}{4}y + 6) < -22\frac{1}{2}$$

× $-2/5$ and REVERSE: $\frac{3}{4}y + 6 > 9$, so $\frac{3}{4}y > 3$

Answer: $y > 4$ one flip, at the first step

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Whatever you do not finish is tonight's practice.
4. The skills for today are posted in Google Classroom.